4A Simplifying Ratios and Dividing in a Ratio

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Simplify each whole-number ratio to its simplest form:

(a) $36 : 54$ (b) $56 : 24$ (c) $75 : 120$ (d) $24 : 36 : 60$

Q2. Simplify each ratio involving decimals, fractions, or mixed units:

(a) $1.8 : 2.4$ (b) $\frac{3}{4} : \frac{5}{6}$ (c) $800\ \text{g} : 2.4\ \text{kg}$ (d) $45\ \text{min} : 2\ \text{hours}$

Q3. Divide each quantity in the given ratio:

(a) $\$140$ in the ratio $3 : 4$ (b) $96\ \text{kg}$ in the ratio $5 : 3$ (c) $250\ \text{mL}$ in the ratio $2 : 3$ (d) $\$360$ in the ratio $2 : 3 : 4$

Q4. Find the unknown value x in each equivalent ratio or scale:

(a) $4 : 7 = 28 : x$ (b) $5 : 9 = x : 63$ (c) Scale $1 : 500$; drawing length $6\ \text{cm}$, real length $x\ \text{m}$ (d) Scale $1 : 200\,000$; map distance $4.5\ \text{cm}$, real distance $x\ \text{km}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A concrete footing mix uses cement, sand, and gravel in the ratio $1 : 2 : 4$ by mass. If $420\ \text{kg}$ of dry concrete mix is prepared, find the mass of (a) cement, (b) sand, and (c) gravel required.

Q6. Two business partners, Noah and Zara, invest in a Melbourne coffee roastery in the ratio $5 : 7$. If Noah invested $\$45\,000$, how much did Zara invest, and what was their combined investment?

Q7. On a Vicmap topographic map with a scale of $1 : 50\,000$, the straight-line distance between two huts in the Alpine National Park is $7.6\ \text{cm}$. Calculate the actual ground distance in kilometres.

Q8. The three interior angles of a triangle are in the ratio $2 : 3 : 4$. Find the size of each angle and classify the triangle by its angles.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. In a fruit punch, the ratio of orange juice to pineapple juice is $3 : 2$, and the ratio of pineapple juice to sparkling water is $4 : 5$. Combine these into a single three-part ratio $\text{orange} : \text{pineapple} : \text{water}$ in simplest form.

Q10. Currently, the ratio of red marbles to blue marbles in a jar is $3 : 5$, and there are 64 marbles in total. How many red marbles must be ADDED to the jar so that the new ratio of red to blue becomes $1 : 1$?